

Feedback — Which Animal Gave Us SARS?

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Thank you. Your submission for this quiz was received.

You submitted this quiz on **Fri 1 May 2015 1:53 PM PDT**. You got a score of **20.00** out of **20.00**.

Question 1

True or False: most distance matrices constructed from biological data are non-additive.

Your Answer	Score	Explanation
☒ True	✓ 2.00	
☐ False		
Total	2.00 / 2.00	

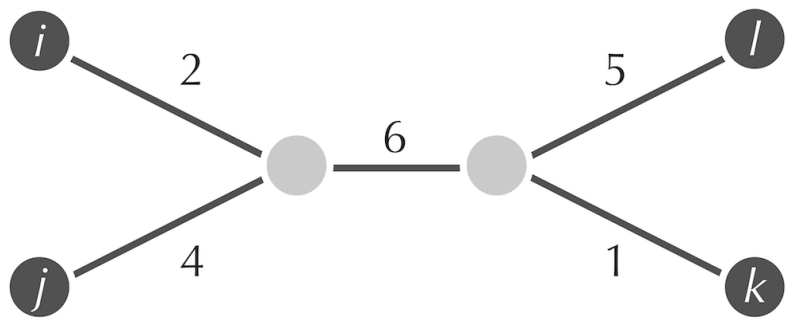
Question 2

True or False: distance-based evolutionary tree construction algorithms can infer ancestral states.

Your Answer	Score	Explanation
☐ True		
☒ False	✓ 2.00	
Total	2.00 / 2.00	

Question 3

Which of the following matrices is fit by the tree shown below?



Your Answer	Score	Explanation
☐ i j k l i 0 6 9 14 j 6 0 11 16 k 9 11 0 7 l 14 16 7 0		
☒ i j k l i 0 6 9 13 j 6 0 11 15 k 9 11 0 6 l 13 15 6 0	✓ 3.00	
☐ i j k l i 0 7 9 13 j 7 0 12 16 k 9 12 0 6 l 13 16 6 0		
☐ i j k l i 0 6 10 13 j 6 0 12 15 k 10 12 0 7 l 13 15 7 0		
Total	3.00 / 3.00	

Question 4

Compute $LlmbLength(k)$ for the additive distance matrix shown below.

i	j	k	l
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```

i  0 14 17 17
j 14  0  7 13
k 17  7  0 16
l 17 13 16  0

```

You entered:

5

Your Answer		Score	Explanation
5	✓	3.00	
Total		3.00 / 3.00	

Question 5

Below is a distance matrix D . Compute $D^*_{k,l}$ where D^* is the neighbor-joining matrix of D .

```

      i  j  k  l
i  0 14 17 17
j 14  0  7 13
k 17  7  0 16
l 17 13 16  0

```

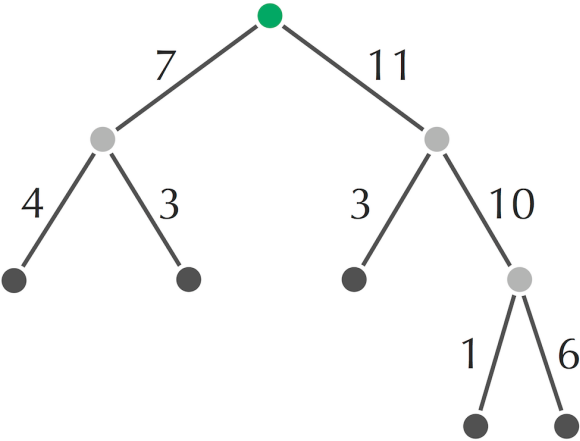
You entered:

-54

Your Answer		Score	Explanation
-54	✓	3.00	
Total		3.00 / 3.00	

Question 6

Compute the parsimony score of the following tree.



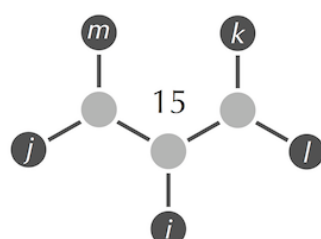
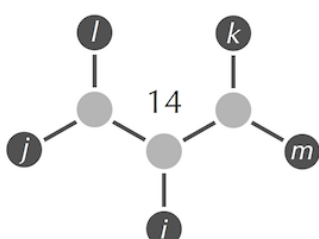
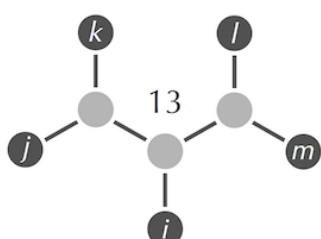
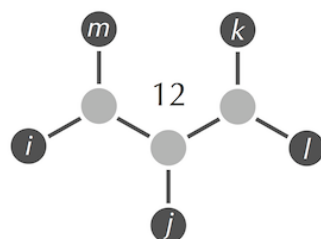
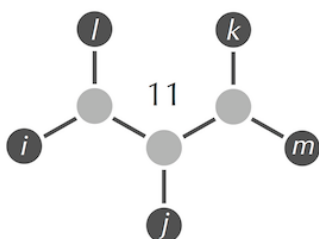
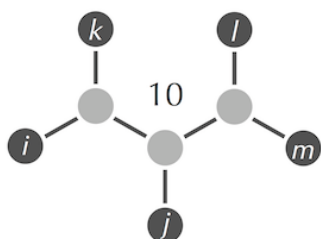
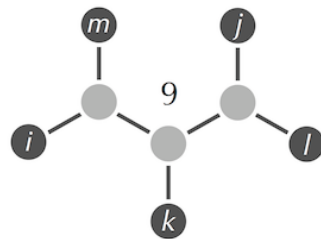
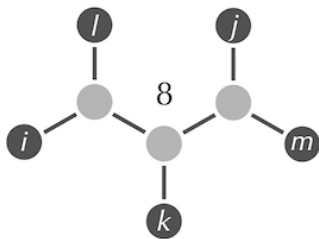
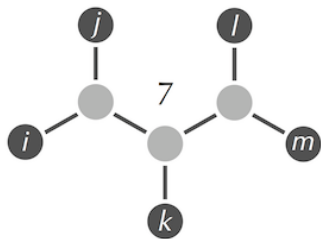
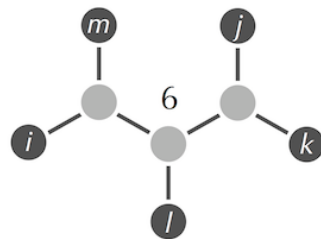
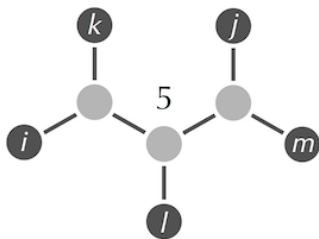
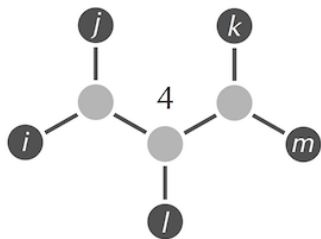
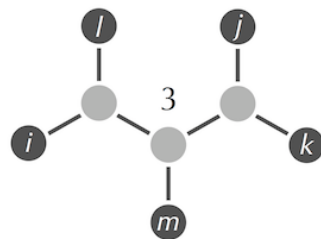
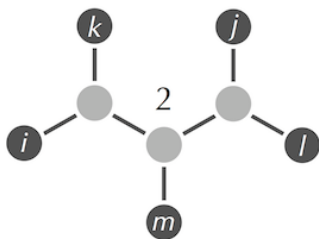
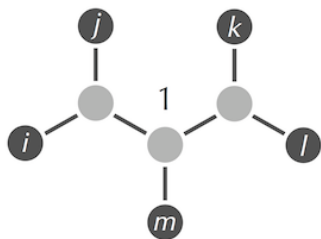
You entered:

45

Your Answer	Score	Explanation
45	✓ 3.00	
Total	3.00 / 3.00	

Question 7

Below are the fifteen unrooted binary trees with five labeled leaves. Which trees are nearest neighbors of tree #1? (Select all that apply.)



Your Answer	Score	Explanation
☒ 15	✓ 0.50	
☐ 8	✓ 0.50	
☐ 3	✓ 0.50	
☒ 12	✓ 0.50	
☒ 7	✓ 0.50	
☒ 4	✓ 0.50	

☐ 5	✓	0.50
☐ 6	✓	0.50
Total		4.00 / 4.00